

Fast Fourier Transform

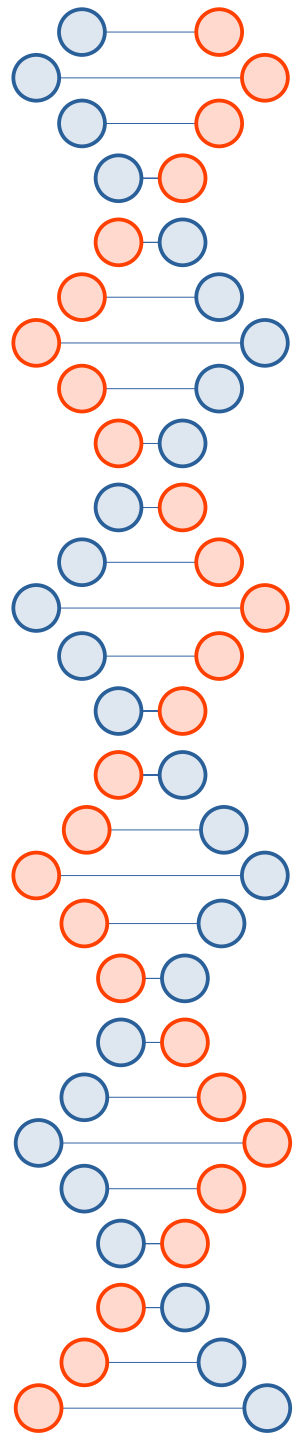
Slides by Siddharth S Ghule,
modified by jedimaestro@asu.edu with help from AI

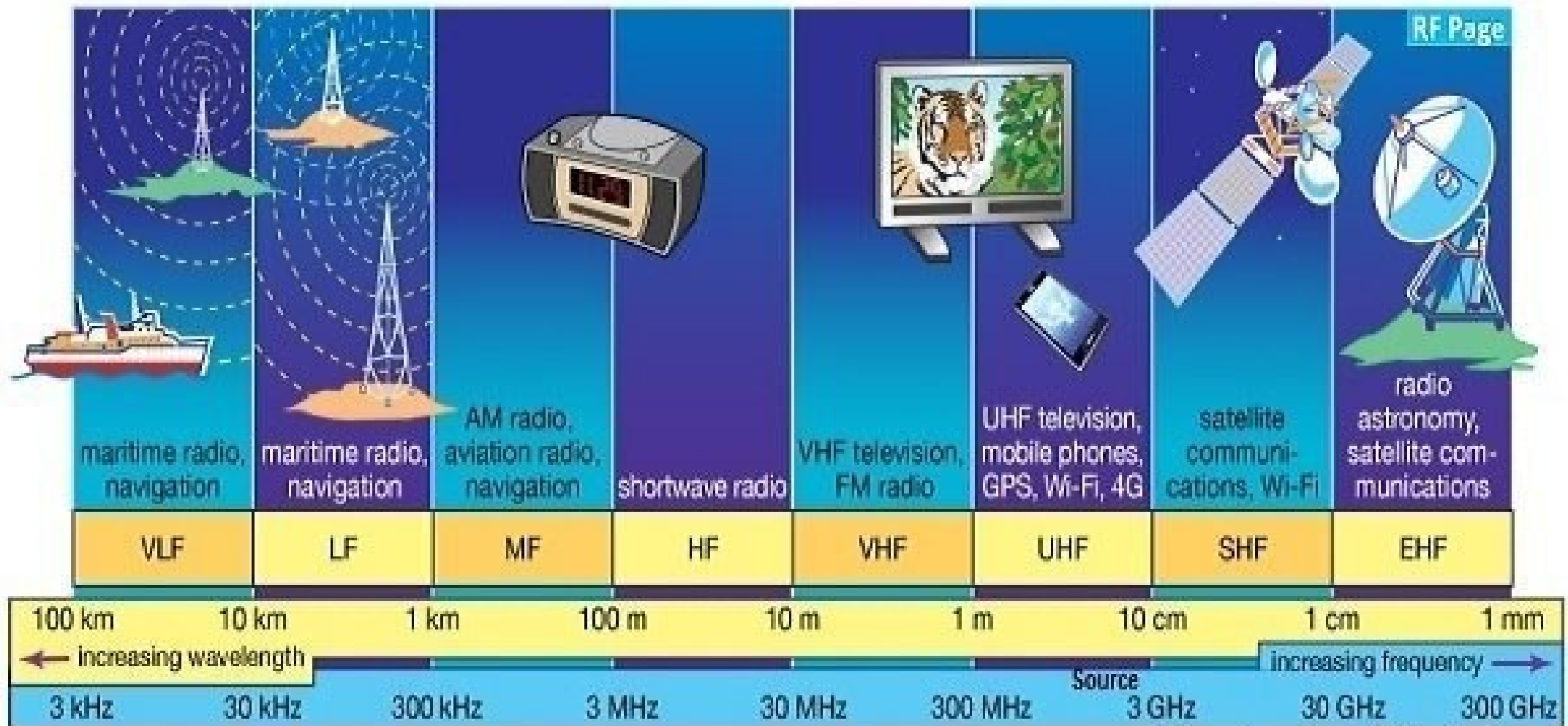
CSE 468 Fall 2026

This next slide is about the first pop quiz, which is coming soon, and is not directly related to FFTs (the first exam will be about FFTs)...

Bitwise XOR as a cipher itself

- Typically used by malware, 8 or 32 bits
 - WEP attack uses these properties
- $(B \text{ xor } K) \text{ xor } K = B$
- $(A \text{ xor } K) \text{ xor } (B \text{ xor } K) = A \text{ xor } B$
- $(0 \text{ xor } K) = K$
- $(K \text{ xor } K) = 0$
- Frequency analysis or brute force





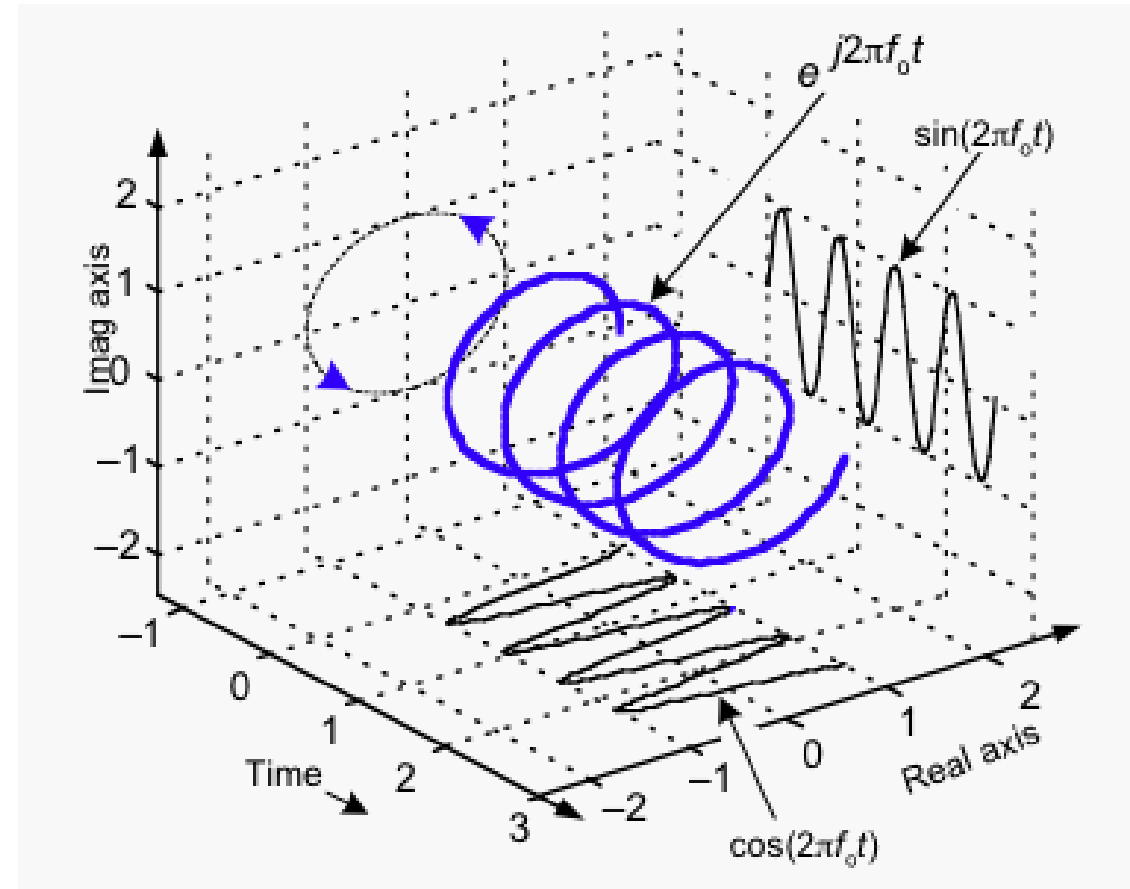
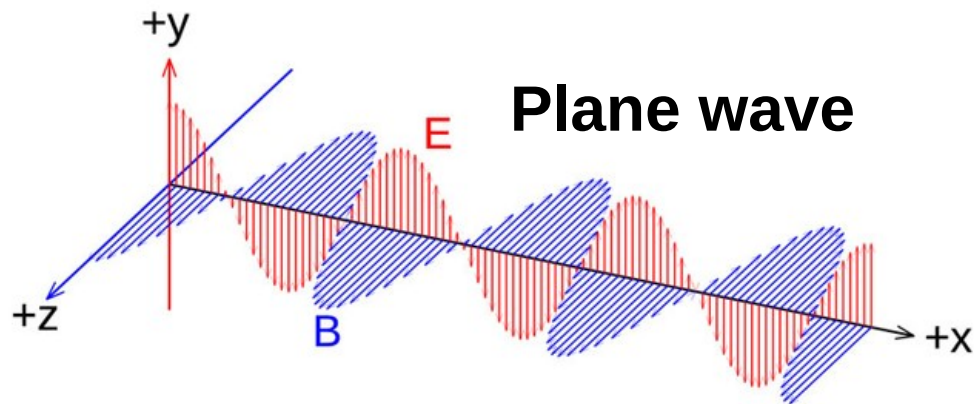
Source: Encyclopaedia Britannica, Inc.

Representing wave as a complex quantity makes math easier.

$$\widetilde{E}(x, t) = E_0 \exp i(\kappa'x - \omega t + \phi)$$

$$E(x, t) = \Re \{ E_0 \exp i(\kappa'x - \omega t + \phi) \}$$

Euler's Theorem: $e^{i\theta} = \cos \theta + i \sin \theta$



<https://www.oceanopticsbook.info/view/theory-electromagnetism/level-2/plane-wave-solutions#x1-1002r1>

<https://math.stackexchange.com/questions/144268/is-there-a-name-for-this-type-of-plot-function-on-complex-plane-vs-time-shown>

[https://www.feynmanlecturesonphysics.com/Bookshelves/Quantum_Mechanics/Introductory_Quantum_Mechanics_\(Fitzpatrick\)/02%3A_Wave-Particle_Duality/2.03%3A_Waves_via_Complex_Functions](https://www.feynmanlecturesonphysics.com/Bookshelves/Quantum_Mechanics/Introductory_Quantum_Mechanics_(Fitzpatrick)/02%3A_Wave-Particle_Duality/2.03%3A_Waves_via_Complex_Functions)

$$\kappa = \frac{2\pi}{\lambda}$$

← Wave number

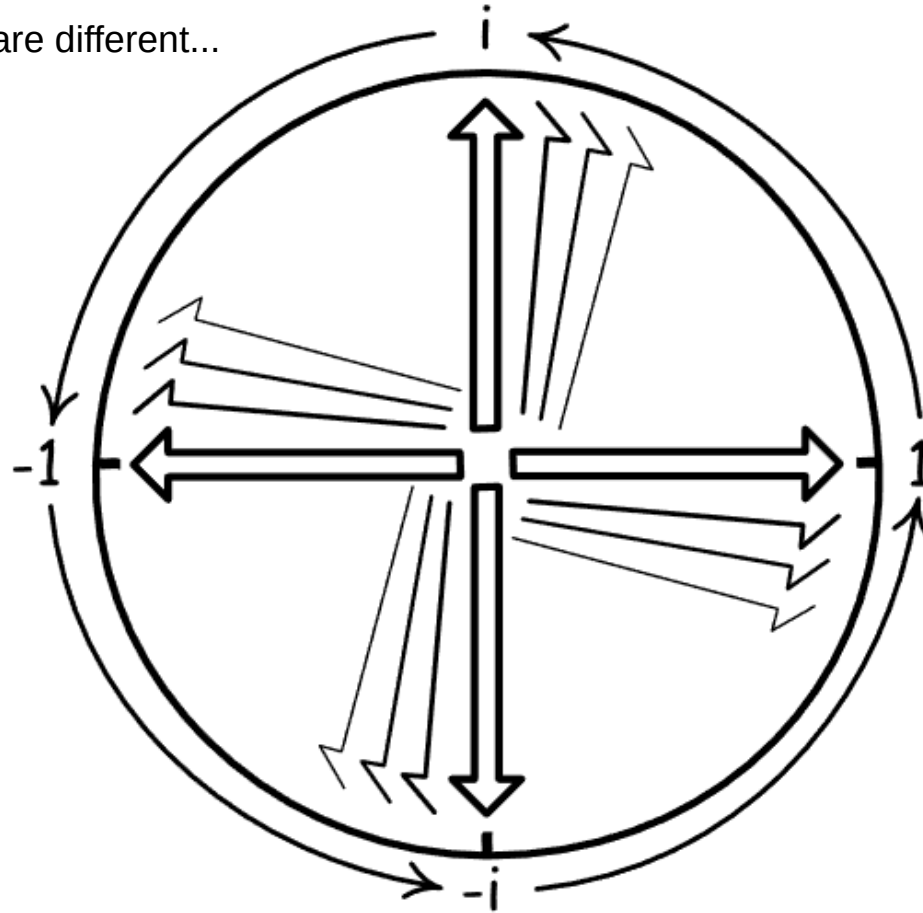
$$\omega = 2\pi\nu = \frac{2\pi}{T}$$

← Angular frequency, ν is the frequency in Hz and T is the period

$$\phi$$

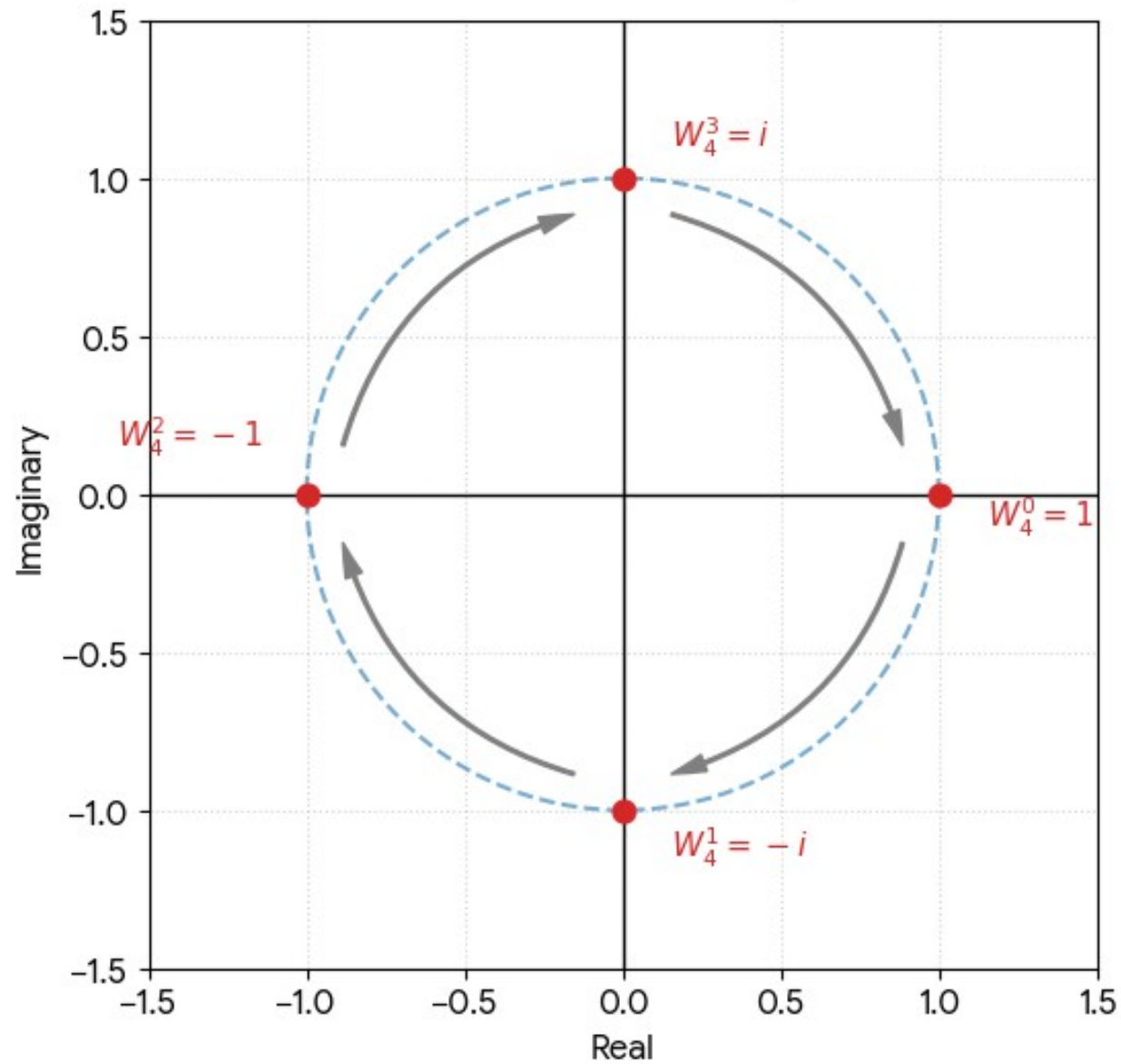
← Phase shift

Complex waves are different...

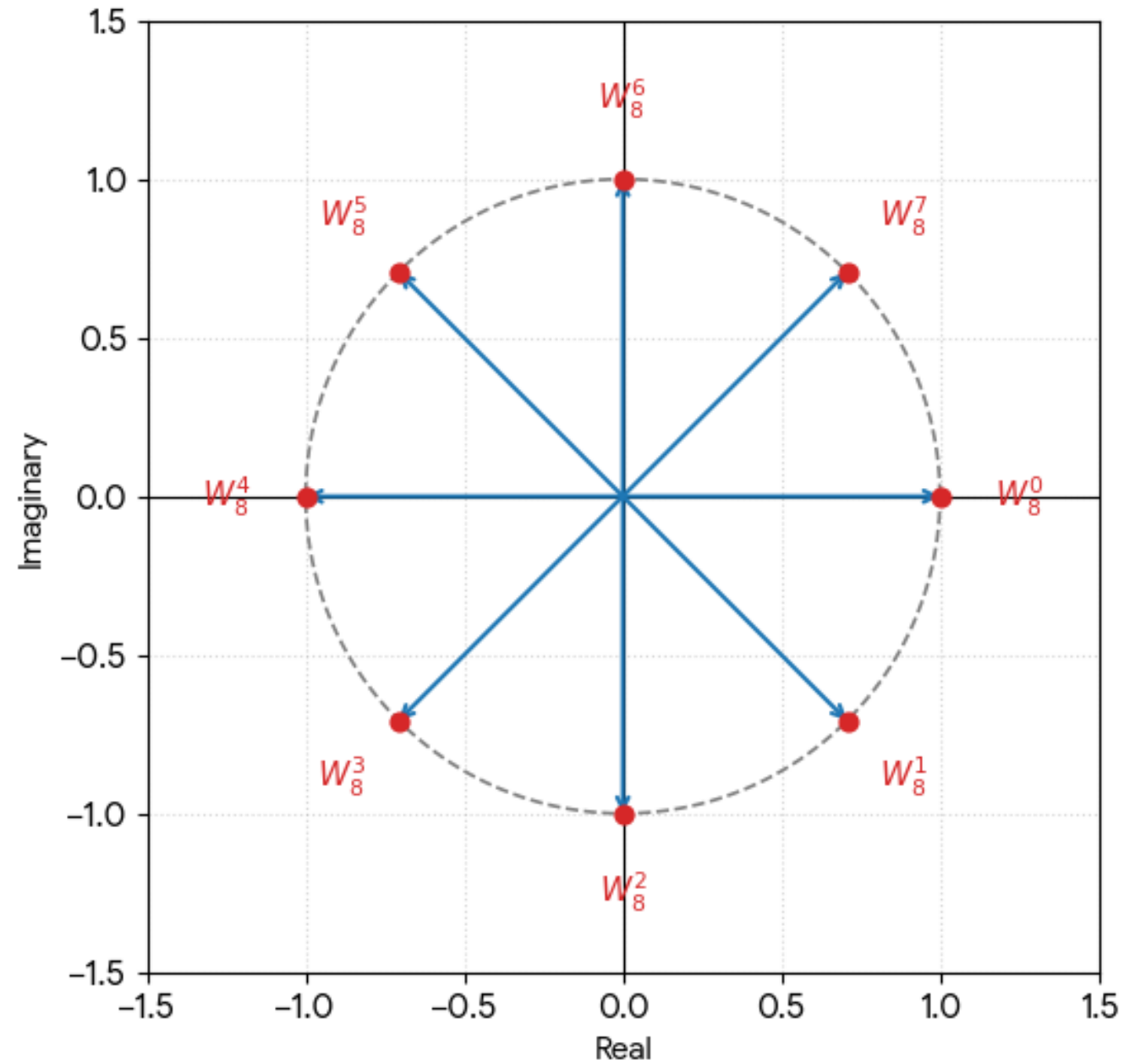


https://sites.pitt.edu/~jdnorton/teaching/HPS_0410/chapters/quantum_theory_complex/complex.html

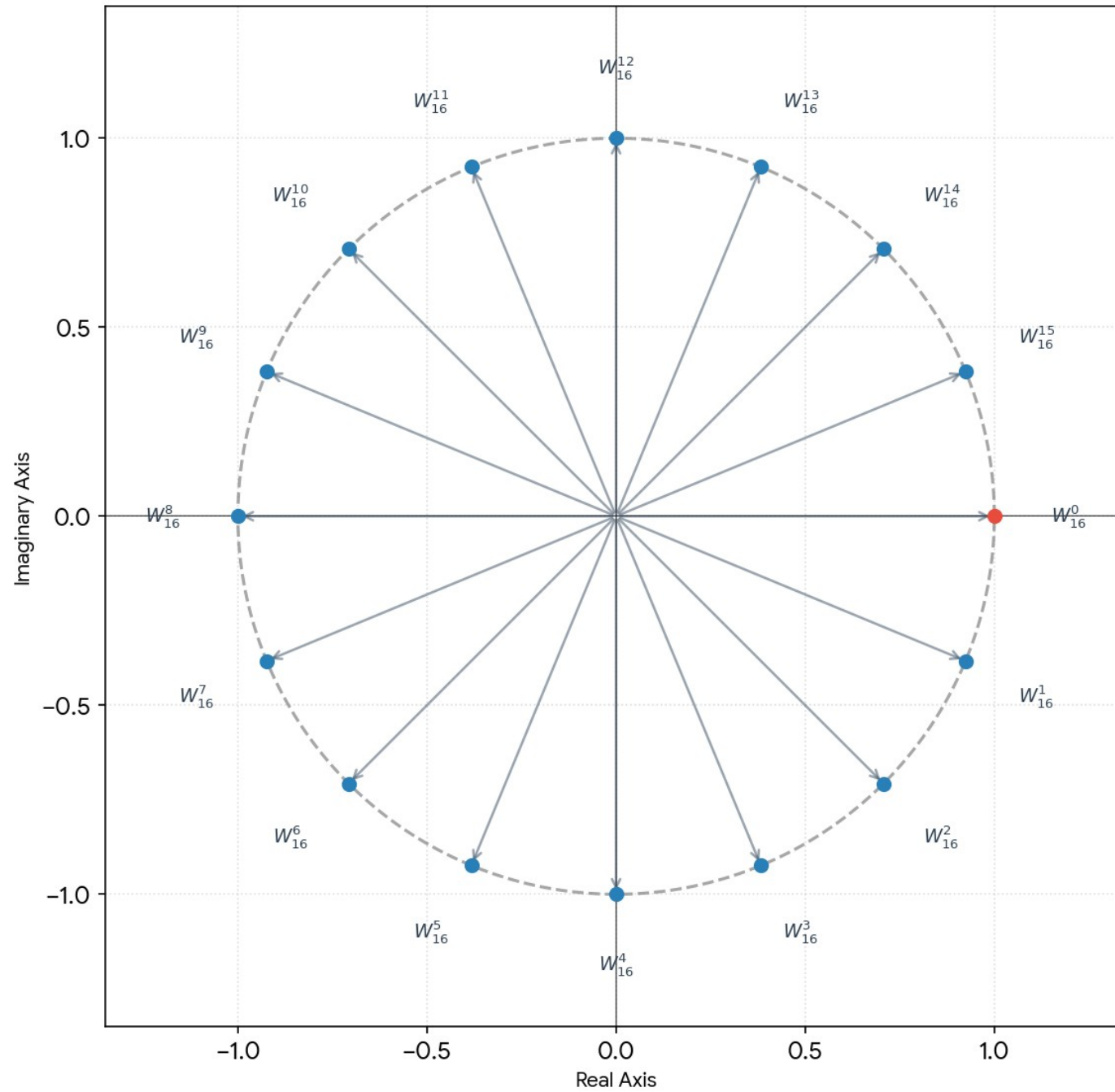
4-Point FFT Twiddle Factors ($W_4^k = e^{-i2\pi k/4}$)



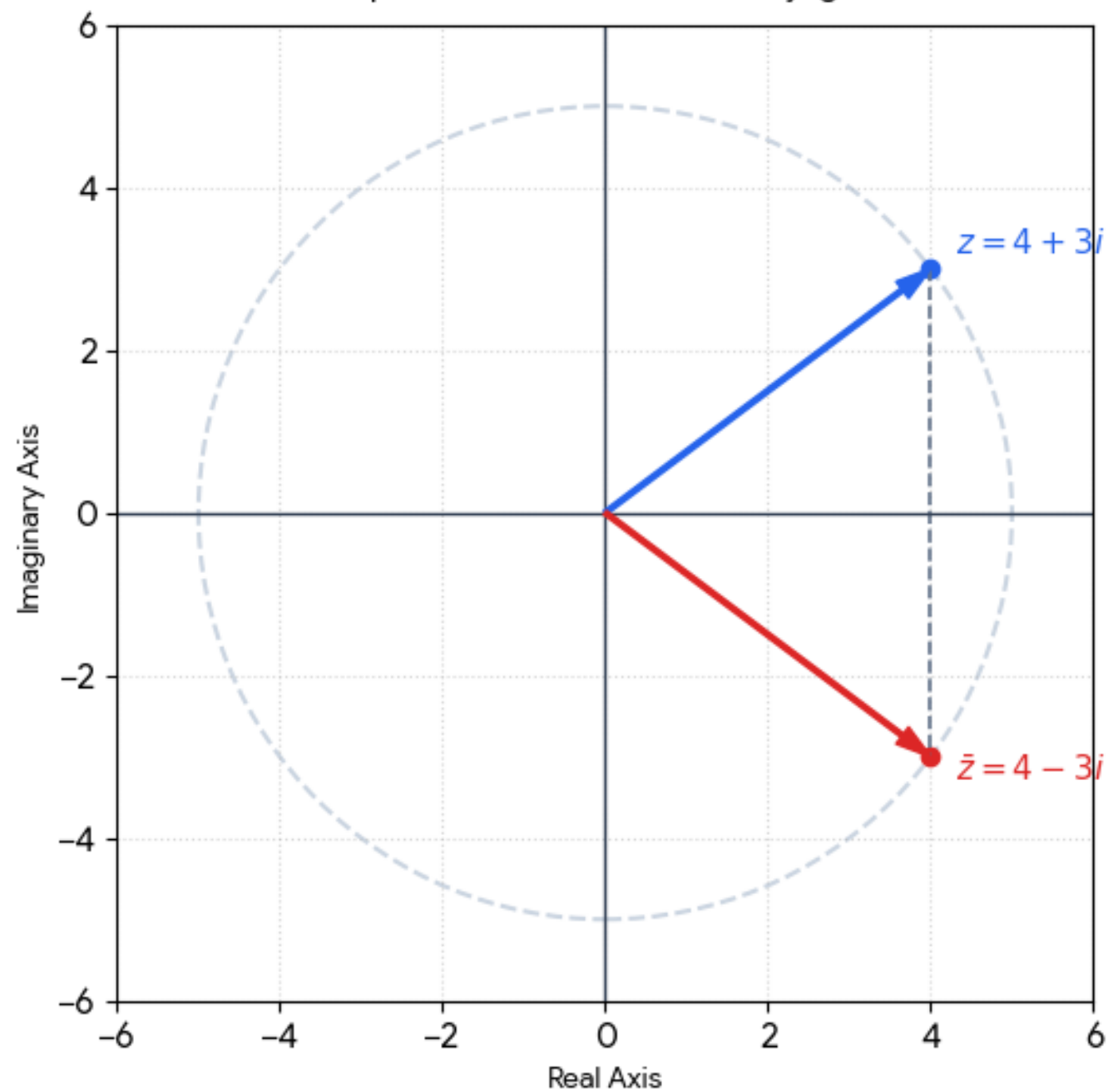
8-Point FFT Twiddle Factors ($W_8^k = e^{-i2\pi k/8}$)



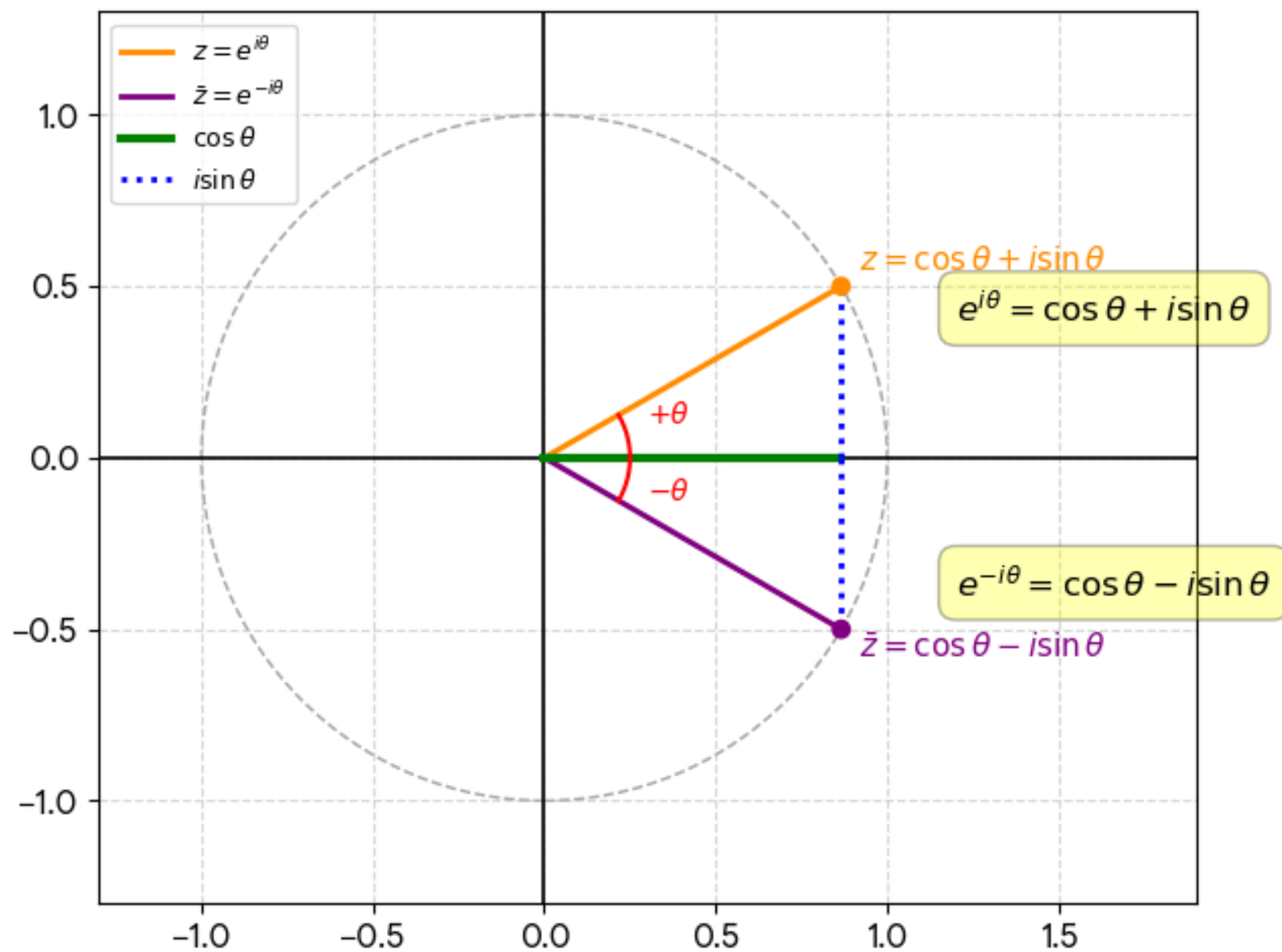
16-Point FFT Twiddle Factors ($W_{16}^k = e^{-i2\pi k/16}$)



Complex Number z and its Conjugate $\bar{z}$



Complex Conjugate and Euler's Formula



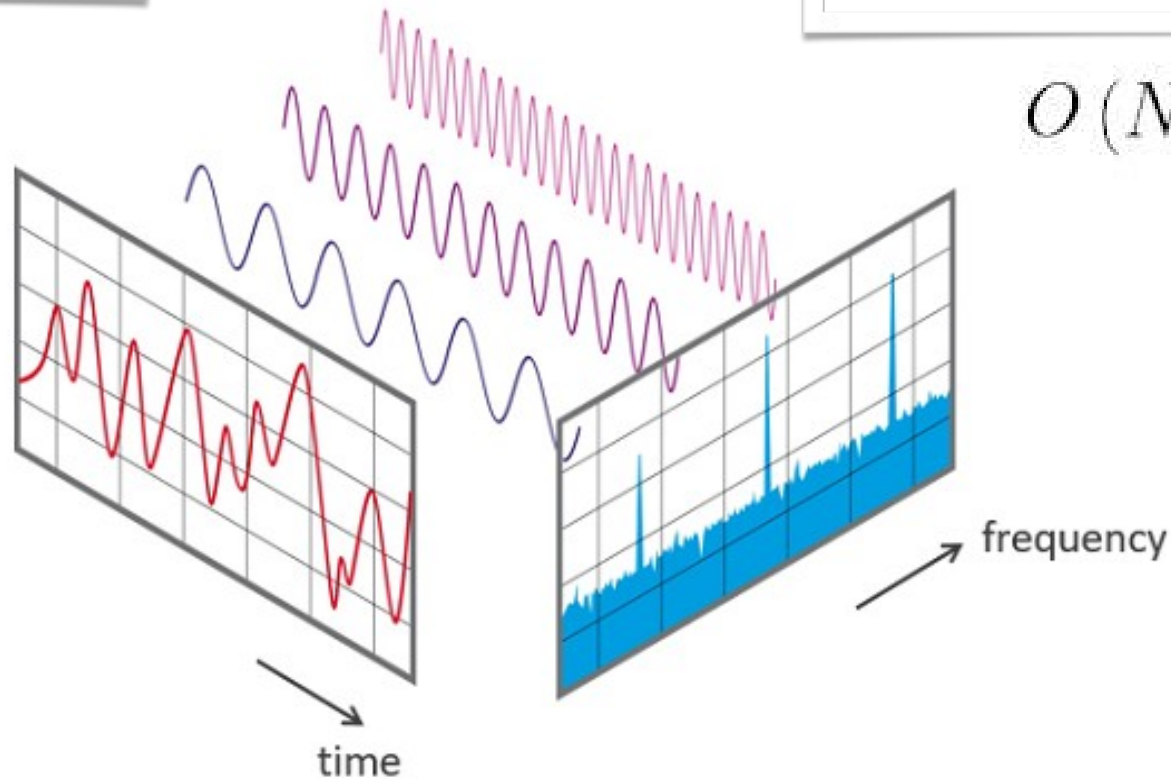
Fourier Transform & Discrete Fourier Transform (DFT)

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$

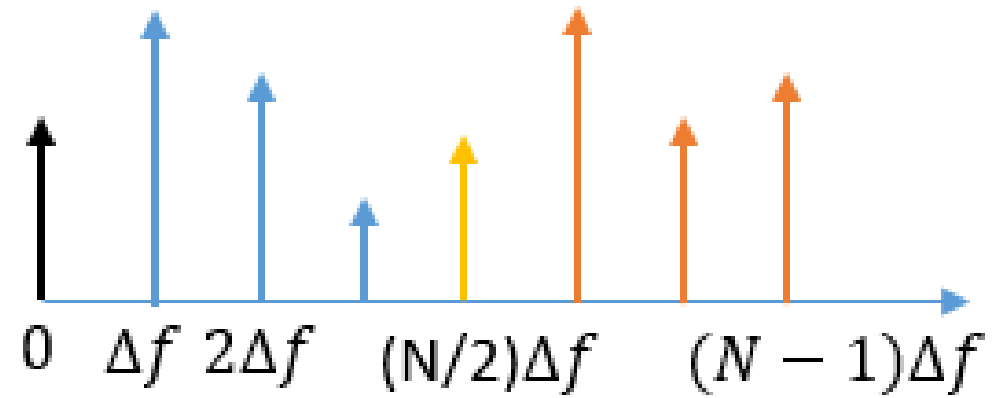
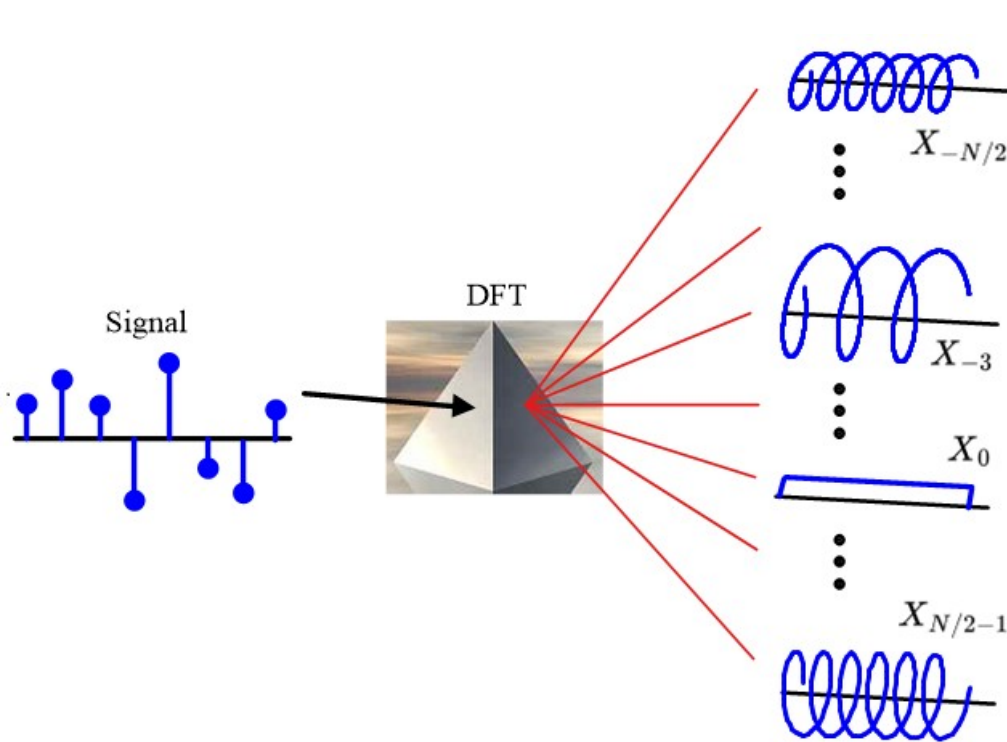
$O(N^2)$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi k n / N}$$

$O(N \log_2 N)$



DFT



$$\Delta f = \frac{F_s}{N}$$

F_s = Sampling frequency

N = Total number of input samples

Frequency Range = $0, \dots, F_s$

$$f_s > 2 \cdot f_{max}$$

Humans can hear frequencies up to 20 kHz

Spotify *etc.* sample music at 44.1 kHz

FFT

Diagram illustrating the Discrete Fourier Transform (DFT) equation and its symmetry property.

The DFT equation is shown as:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}$$

Annotations for the DFT equation:

- $N-1$: Total Number of Samples (indicated by a red arrow)
- $X[k]$: Frequency bins (indicated by a green arrow)
- $x[n]$: Input Samples (indicated by a purple arrow)
- W_N^{nk} : Twiddle Factor (indicated by a blue arrow)

The symmetry property is shown as:

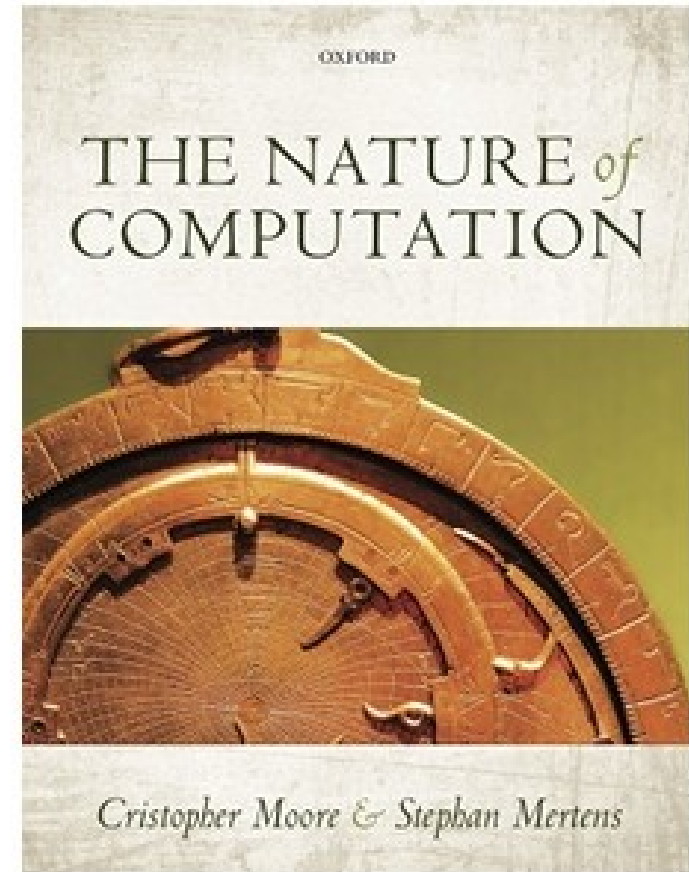
$$X[k] = E[k] + \omega_k \cdot O[k]$$
$$X[k + \frac{N}{2}] = E[k] - \omega_k \cdot O[k]$$

The range of k is specified as:

$$k = 0 \text{ to } \frac{N}{2} - 1$$

$$\begin{pmatrix} f(0) \\ f(1) \\ f(2) \\ \vdots \end{pmatrix} = \frac{1}{\sqrt{n}} \begin{pmatrix} 1 & 1 & 1 & \dots \\ 1 & \omega_n & \omega_n^2 & \dots \\ 1 & \omega_n^2 & \omega_n^4 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \cdot \begin{pmatrix} \tilde{f}(0) \\ \tilde{f}(1) \\ \tilde{f}(2) \\ \vdots \end{pmatrix}$$

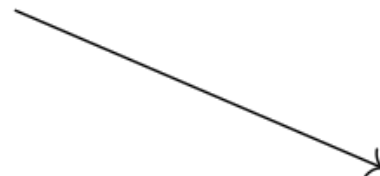
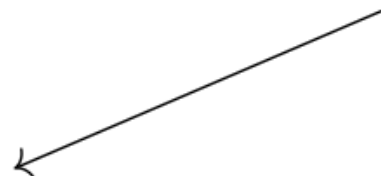
$$f = Q \cdot \tilde{f} \text{ where } Q_{tk} = \frac{1}{\sqrt{n}} \omega_n^{kt}.$$



<https://nature-of-computation.org/>

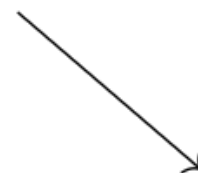
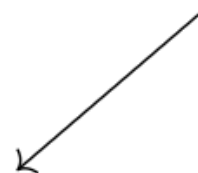
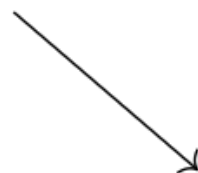
$$[-12 + 9i, -11 - 2i, 8 - 14i, -5 - 12i, 11 + 1i, 13 + 9i, 13 - 2i, -10 + 4i]$$

$$[-12 + 9i, -11 - 2i, 8 - 14i, -5 - 12i, 11 + 1i, 13 + 9i, 13 - 2i, -10 + 4i]$$



$$[-12 + 9i, 8 - 14i, 11 + 1i, 13 - 2i]$$

$$[-11 - 2i, -5 - 12i, 13 + 9i, -10 + 4i]$$



$$[-12 + 9i, 11 + 1i]$$

$$[8 - 14i, 13 - 2i]$$

$$[-11 - 2i, 13 + 9i]$$

$$[-5 - 12i, -10 + 4i]$$



$$[\tilde{x}1_{ev, ev, ev}, \tilde{x}1_{ev, ev, od}]$$

$$[\tilde{x}1_{ev, od, ev}, \tilde{x}1_{ev, od, od}]$$

$$[\tilde{x}1_{od, ev, ev}, \tilde{x}1_{od, ev, od}]$$

$$[\tilde{x}1_{od, od, ev}, \tilde{x}1_{od, od, od}]$$

$$[-12 + 9i, -11 - 2i, 8 - 14i, -5 - 12i, 11 + 1i, 13 + 9i, 13 - 2i, -10 + 4i]$$

$$[-12 + 9i, 8 - 14i, 11 + 1i, 13 - 2i]$$

$$[-11 - 2i, -5 - 12i, 13 + 9i, -10 + 4i]$$

$$[-12 + 9i, 11 + 1i]$$

$$[8 - 14i, 13 - 2i]$$

$$[-11 - 2i, 13 + 9i]$$

$$[-5 - 12i, -10 + 4i]$$

$$[\tilde{x}1_{ev, ev, ev}, \tilde{x}1_{ev, ev, od}]$$

$$[\tilde{x}1_{ev, od, ev}, \tilde{x}1_{ev, od, od}]$$

$$[\tilde{x}1_{od, ev, ev}, \tilde{x}1_{od, ev, od}]$$

$$[\tilde{x}1_{od, od, ev}, \tilde{x}1_{od, od, od}]$$

Twiddle Factors

$$\omega_k = e^{-2\pi i k/2}$$

$$\omega_0 : e^{-2\pi i \cdot 0/8} = e^0 = 1$$

$$\omega_1 : e^{-2\pi i \cdot 1/8} = e^{-\pi i/4} = \cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right) = \frac{1-i}{\sqrt{2}}$$

$$\omega_2 : e^{-2\pi i \cdot 2/8} = e^{-\pi i/2} = \cos\left(\frac{-\pi}{2}\right) + i \sin\left(\frac{-\pi}{2}\right) = -i$$

$$\omega_3 : e^{-2\pi i \cdot 3/8} = e^{-\pi i 3/4} = \cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right) = -\frac{1+i}{\sqrt{2}}$$

What is the 1-point FFT of each of these samples?

- $x = [4]$
- $x = [-16]$
- $x = [7 + 5i]$
- $x = [9 - 2i]$

What is the 1-point FFT of each of these samples?

- $x = [4] \rightarrow X = [4]$
- $x = [-16] \rightarrow X = [-16]$
- $x = [7 + 5i] \rightarrow X = [7 + 5i]$
- $x = [9 - 2i] \rightarrow X = [9 - 2i]$

2-point FFT of $x = [7, 16]$

$$W_2^0 = e^{-i2\pi(0)/2} = 1$$

$$X(0) = x(0) + W_2^0 \cdot x(1)$$

$$X(1) = x(0) - W_2^0 \cdot x(1)$$

$$X(0) = 7 + 1 \cdot 16 = 7 + 16 = 23$$

$$X(1) = 7 - 1 \cdot 16 = 7 - 16 = -9$$

Answer is $X = [23, -9]$

$$x = [7 + 6i, 3 - 2i, 4 - 3i, -5 - i]$$

$$E(0) = x(0) + x(2) = (7 + 6i) + (4 - 3i) = 11 + 3i$$

$$E(1) = x(0) - x(2) = (7 + 6i) - (4 - 3i) = 3 + 9i$$

$$O(0) = x(1) + x(3) = (3 - 2i) + (-5 - i) = -2 - 3i$$

$$O(1) = x(1) - x(3) = (3 - 2i) - (-5 - i) = 8 - i$$

$$W_4^0 = 1 \text{ and } W_4^1 = -i.$$

$$X(0) = E(0) + W_4^0 \cdot O(0) = (11 + 3i) + 1 \cdot (-2 - 3i) = 9$$

$$X(1) = E(1) + W_4^1 \cdot O(1) = (3 + 9i) + (-i) \cdot (8 - i) = 2 + i$$

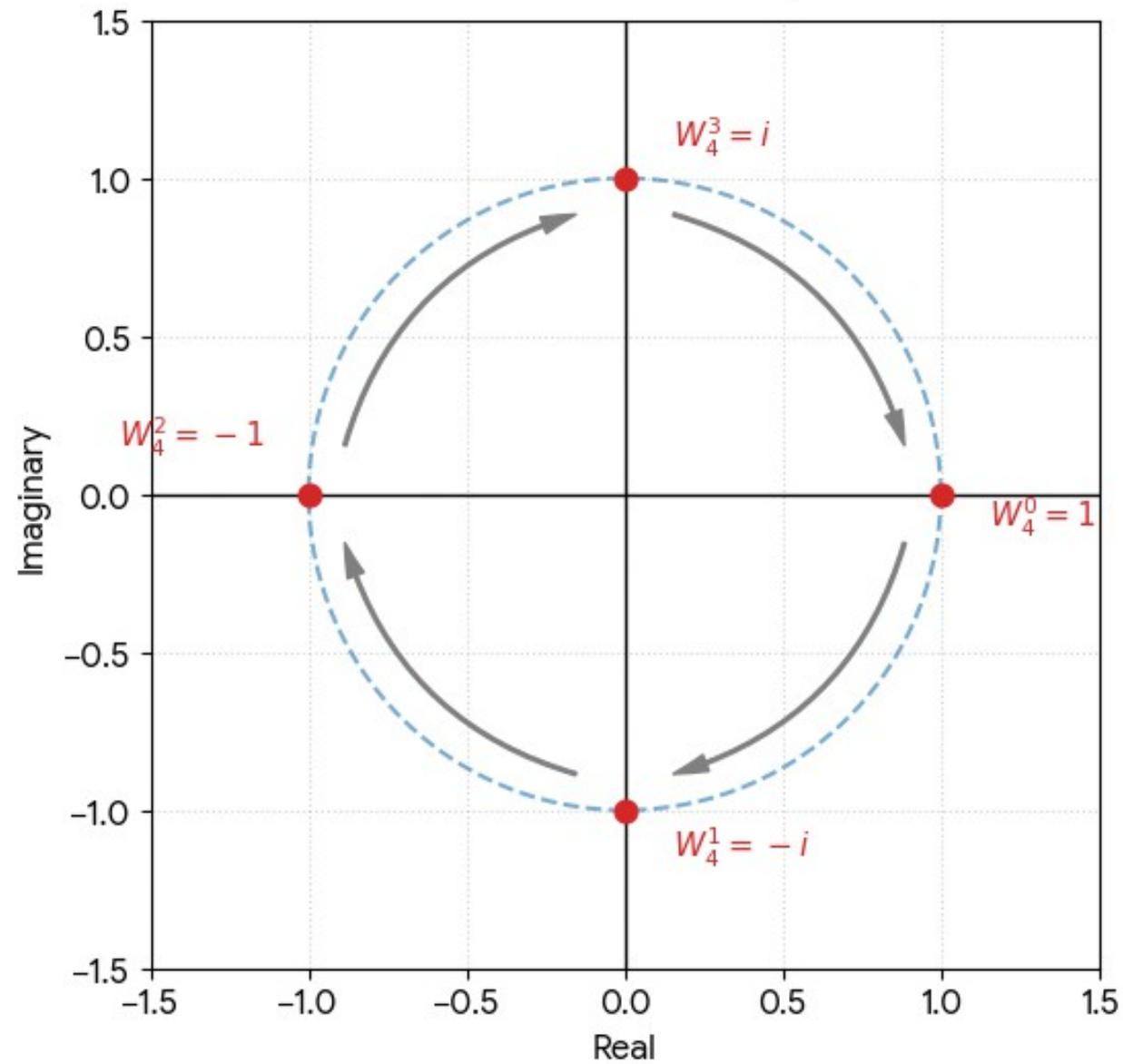
$$X(2) = E(0) - W_4^0 \cdot O(0) = (11 + 3i) - 1 \cdot (-2 - 3i) = 13 + 6i$$

$$X(3) = E(1) - W_4^1 \cdot O(1) = (3 + 9i) - (-i) \cdot (8 - i) = 4 + 17i$$

$$x = [7 + 6i, 3 - 2i, 4 - 3i, -5 - i]$$

$$X = [9, 2 + i, 13 + 6i, 4 + 17i]$$

4-Point FFT Twiddle Factors ($W_4^k = e^{-i2\pi k/4}$)



$$\mathbf{x} = [1, 8, -3, 2.9]$$

$$E(0) = x(0) + x(2) = 1 + (-3) = -2$$

$$E(1) = x(0) - x(2) = 1 - (-3) = 4$$

$$O(0) = x(1) + x(3) = 8 + 2.9 = 10.9$$

$$O(1) = x(1) - x(3) = 8 - 2.9 = 5.1$$

$$W_4^0 = 1 \text{ and } W_4^1 = -i.$$

$$X(0) = E(0) + W_4^0 \cdot O(0) = -2 + 1 \cdot 10.9 = 8.9$$

$$X(1) = E(1) + W_4^1 \cdot O(1) = 4 + (-i) \cdot 5.1 = 4 - 5.1i$$

$$X(2) = E(0) - W_4^0 \cdot O(0) = -2 - 1 \cdot 10.9 = -12.9$$

$$X(3) = E(1) - W_4^1 \cdot O(1) = 4 - (-i) \cdot 5.1 = 4 + 5.1i$$

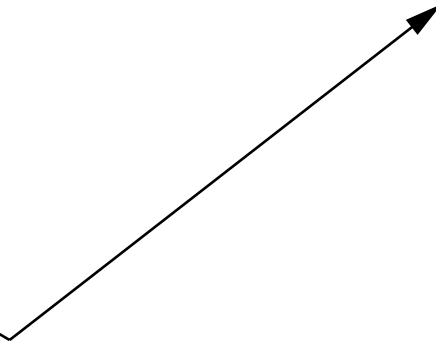
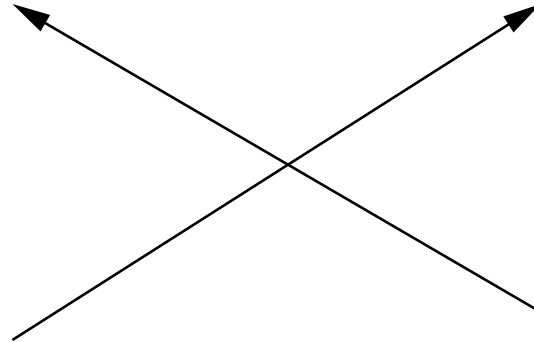
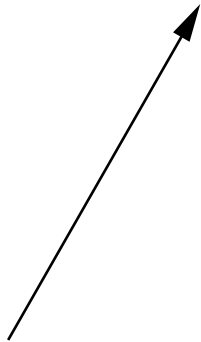
$$\mathbf{x} = [1, 8, -3, 2.9]$$

$$\mathbf{X} = [8.9, 4 - 5.1i, -12.9, 4 + 5.1i]$$

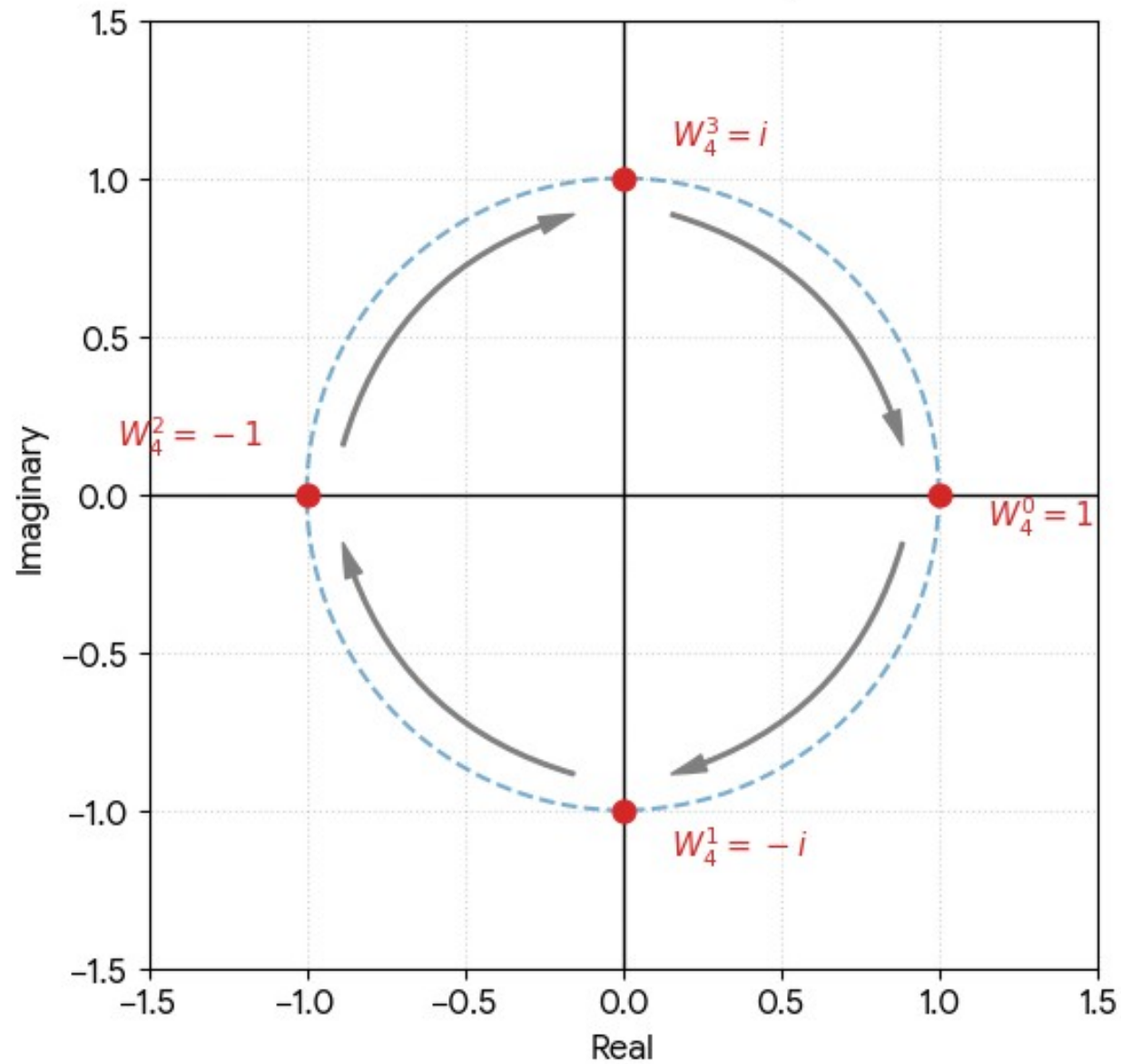
DC component

Nyquist frequency

Complex conjugates



4-Point FFT Twiddle Factors ($W_4^k = e^{-i2\pi k/4}$)

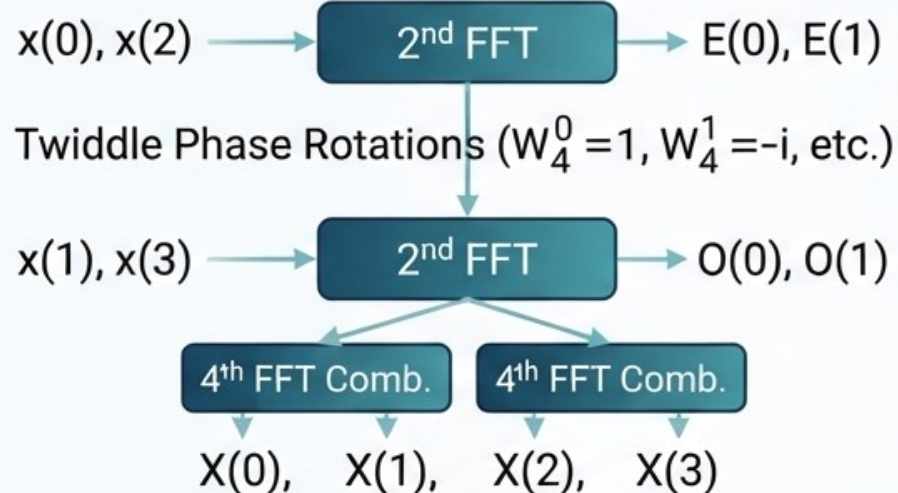




Looking Ahead: The Quantum Fourier Transform (QFT)

Classical 4-Point FFT (Butterfly Network)

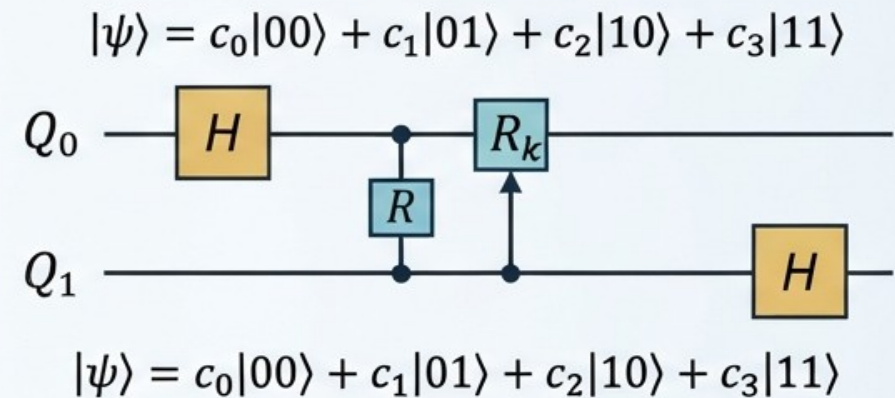
(Butterfly Network)



Input Array Size: 4 elements = 4 Dimensions.
Data: Complex Time-Domain Samples.

2-Qubit Quantum State Space (Entangled)

(Entangled)



Qubit Space Size: $2^2 = 4$ Dimensions.
Data: Complex Probability Amplitudes.


**4 Dimensions
= 2 Qubits:
Native
Exponential
Scaling**



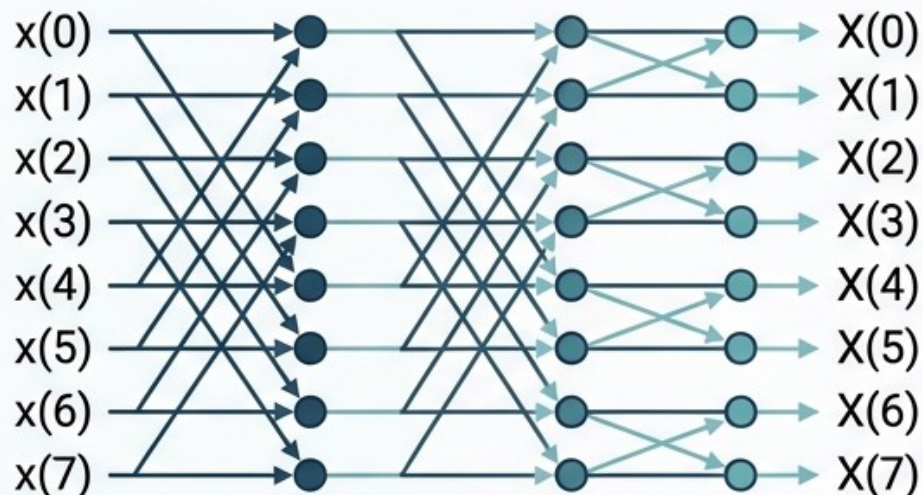
Preview for Week 15: The Exponential Advantage

- Hardware Parallels:** Classical butterfly network operations (Sum/Difference) and twiddle factor rotations map directly to fundamental Quantum Gates (Hadamard, Controlled-Phase).
- Computational Speedup:** Running a native butterfly network inside exponentially scalable entangled states allows the QFT (Quantum Fourier Transform) to achieve an exponential speedup over classical FFT algorithms (e.g., Shor's Algorithm (e.g., Shor's Algorithm)).

Looking Ahead: The Quantum Fourier Transform (QFT)

Classical 8-Point FFT (Butterfly Network)

(Butterfly Network)



Input Array Size: 8 elements = 8 Dimensions.

Data: Complex Time-Domain Samples.

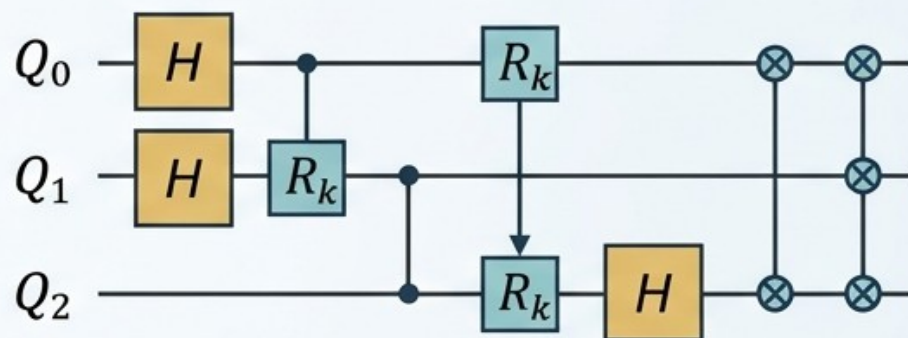


**8 Dimensions
= 3 Qubits:
Native
Exponential
Scaling**

3-Qubit Quantum State Space (Entangled)

(Entangled)

$$|\psi\rangle = c_0|000\rangle + c_1|001\rangle + \dots + c_7|111\rangle$$



$$|\psi\rangle = c_0|000\rangle + c_1|001\rangle + \dots + c_7|111\rangle$$

Qubit Space Size: $2^3 = 8$ Dimensions.

Data: Complex Probability Amplitudes.



Preview for Week 15: The Exponential Advantage

1. **Hardware Parallels:** Classical butterfly network operations (Sum/Difference) and twiddle factor rotations map directly to fundamental Quantum Gates (Hadamard, Controlled-Phase).
2. **Computational Speedup:** Running a native butterfly network inside exponentially scalable entangled states allows the QFT (Quantum Fourier Transform) to achieve an exponential speedup over classical FFT algorithms (e.g., Shor's Algorithm (e.g., Shor's Algorithm)).

How to prepare for the FFT on the exam and pop quiz

- Check Canvas, I'll post example exams and solutions
- *The Nature of Computation*, by Cris Moore and Stephan Mertens, pages 49–53
- Tons of YouTube videos



This presentation was created
with the collaborative
assistance of Google Gemini.

[We promise we validated the fingers.]





This presentation was created
with the collaborative
assistance of Google Gemini.

[But we think we hallucinated an N=3 butterfly
and a fractional twiddle factor in the fingers!]

